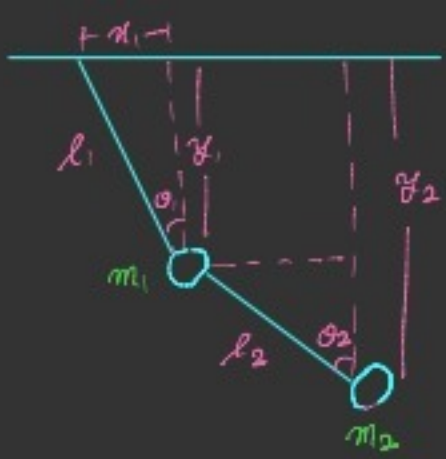




$$x_1 = l_1 \sin(\theta_1)$$

$$y_1 = -l_1 \cos(\theta_1)$$

$$x_2 = x_1 + l_2 \sin(\theta_2) = l_1 \sin(\theta_1) + l_2 \sin(\theta_2)$$

$$y_2 = y_1 - l_2 \cos(\theta_2) = -l_1 \cos(\theta_1) - l_2 \cos(\theta_2)$$

$$\dot{x}_2 = l_1 \cos(\theta_1) \dot{\theta}_1 + l_2 \cos(\theta_2) \dot{\theta}_2 = v_x$$

$$\dot{y}_2 = l_1 \sin(\theta_1) \dot{\theta}_1 + l_2 \sin(\theta_2) \dot{\theta}_2 = v_y$$

$$l_1 \sin(\theta_1) \dot{\theta}_1 + l_2 \sin(\theta_2) \dot{\theta}_2$$

$$v_{\text{net}} = \sqrt{v_x^2 + v_y^2}$$

$$\begin{aligned} v_{\text{net}}^2 &= l_1^2 \cos^2(\theta_1) \dot{\theta}_1^2 + l_2^2 \cos^2(\theta_2) \dot{\theta}_2^2 + 2 l_1 l_2 \cos(\theta_1) \cos(\theta_2) \dot{\theta}_1 \dot{\theta}_2 + l_1^2 \sin^2(\theta_1) \dot{\theta}_1^2 + l_2^2 \sin^2(\theta_2) \dot{\theta}_2^2 + 2 l_1 l_2 \sin(\theta_1) \sin(\theta_2) \dot{\theta}_1 \dot{\theta}_2 \\ &= l_1^2 \dot{\theta}_1^2 + l_2^2 \dot{\theta}_2^2 + 2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 (\cos(\theta_1) \cos(\theta_2) + \sin(\theta_1) \sin(\theta_2)) \end{aligned}$$

$$v_{\text{net}}^2 = l_1^2 \dot{\theta}_1^2 + l_2^2 \dot{\theta}_2^2 + 2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 (\cos(\theta_1 - \theta_2))$$

$$T_2 = \frac{1}{2} m_2 v^2 = \frac{1}{2} m_2 (l_1^2 \dot{\theta}_1^2 + l_2^2 \dot{\theta}_2^2 + 2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_1 - \theta_2))$$

$$V_2 = m_2 g h = -m_2 g (l_1 \cos(\theta_1) + l_2 \cos(\theta_2))$$

$$L_2 = T_2 - V_2 = m_2 \left[\frac{1}{2} (l_1^2 \dot{\theta}_1^2 + l_2^2 \dot{\theta}_2^2 + 2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_1 - \theta_2)) + g (l_1 \cos(\theta_1) + l_2 \cos(\theta_2)) \right]$$

$$\text{Now: } L = L_1 + L_2$$

$$L = \frac{1}{2} m_1 l^2 \dot{\theta}_1^2 + m_1 g l \cos(\theta_1) + m_2 \left[\frac{1}{2} (l_1^2 \dot{\theta}_1^2 + l_2^2 \dot{\theta}_2^2 + 2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_1 - \theta_2)) + g (l_1 \cos(\theta_1) + l_2 \cos(\theta_2)) \right]$$

✖ for m_2 :

$$i) \frac{\partial L}{\partial \dot{\theta}_2} = \frac{m_2}{2} \left(l_2^2 \dot{\theta}_2 + 2l_1 l_2 \dot{\theta}_1 \cos(\theta_1 - \theta_2) \right)$$

$$ii) \frac{d}{dt} = m_2 \left(l_2^2 \ddot{\theta}_2 + l_1 l_2 (\ddot{\theta}_1 \cos(\theta_1 - \theta_2) - \dot{\theta}_1 \sin(\theta_1 - \theta_2) (\dot{\theta}_1 - \dot{\theta}_2)) \right)$$

$$iii) \frac{\partial L}{\partial \theta_2} = -\frac{m_2}{2} \left[2l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2) (-1) \right] - m_2 g l_2 \sin(\theta_2)$$

$$iv) l_2^2 \ddot{\theta}_2 + l_1 l_2 [\ddot{\theta}_1 \cos(\theta_1 - \theta_2) - \dot{\theta}_1 \sin(\theta_1 - \theta_2) (\dot{\theta}_1 - \dot{\theta}_2)] + \frac{1}{2} [2l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2) (-1)] = -g l_2 \sin(\theta_2)$$

$$\Rightarrow l_2^2 \ddot{\theta}_2 = -l_1 l_2 [\ddot{\theta}_1 \cos(\theta_1 - \theta_2) - \dot{\theta}_1 \sin(\theta_1 - \theta_2) (\dot{\theta}_1 - \dot{\theta}_2)] - [l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2) (-1)] - g l_2 \sin(\theta_2)$$

$$\Rightarrow l_2 \ddot{\theta}_2 = -l_1 [\ddot{\theta}_1 \cos(\theta_1 - \theta_2) - \dot{\theta}_1 \sin(\theta_1 - \theta_2) (\dot{\theta}_1 - \dot{\theta}_2)] + [l_1 \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2)] - g \sin(\theta_2)$$

$$= -l_1 \ddot{\theta}_1 \cos(\theta_1 - \theta_2) + \underline{l_1 \dot{\theta}_1 \sin(\theta_1 - \theta_2) (\dot{\theta}_1 - \dot{\theta}_2)} + \underline{l_1 \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2)} - g \sin(\theta_2)$$

$$= -l_1 \ddot{\theta}_1 \cos(\theta_1 - \theta_2) + l_1 \dot{\theta}_1 \sin(\theta_1 - \theta_2) [\dot{\theta}_1 - \dot{\theta}_2 + \dot{\theta}_2] - g \sin(\theta_2)$$

$$= -l_1 \ddot{\theta}_1 \cos(\theta_1 - \theta_2) + l_1 \dot{\theta}_1^2 \sin(\theta_1 - \theta_2) - g \sin(\theta_2)$$

$$\Rightarrow \ddot{\theta}_2 = -\frac{l_1}{l_2} \ddot{\theta}_1 \cos(\theta_1 - \theta_2) + \frac{l_1 \dot{\theta}_1^2 \sin(\theta_1 - \theta_2)}{l_2} - \frac{g \sin(\theta_2)}{l_2}$$

* for m_1 :

$$L = \frac{1}{2} m_1 l_1^2 \dot{\theta}_1^2 + m_1 g l_1 \cos(\theta_1) + m_2 \left[\frac{1}{2} (l_1^2 \dot{\theta}_1^2 + l_2^2 \dot{\theta}_2^2 + 2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_1 - \theta_2)) + g (l_1 \cos(\theta_1) + l_2 \cos(\theta_2)) \right]$$

$$i) \quad \frac{\partial L}{\partial \dot{\theta}_1} = m_1 l_1^2 \dot{\theta}_1 + m_2 [l_1^2 \dot{\theta}_1 + l_1 l_2 \dot{\theta}_2 \cos(\theta_1 - \theta_2)]$$

$$ii) \quad \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}_1} \right) = m_1 l_1^2 \ddot{\theta}_1 + m_2 [l_1^2 \ddot{\theta}_1 + l_1 l_2 \{ \ddot{\theta}_2 \cos(\theta_1 - \theta_2) - \dot{\theta}_2 \sin(\theta_1 - \theta_2) (\dot{\theta}_1 - \dot{\theta}_2) \}]$$

$$iii) \quad \frac{\partial L}{\partial \theta_1} = -m_1 g l_1 \sin(\theta_1) - m_2 [l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2) + g l_1 \sin(\theta_1)]$$

$$iv) \quad m_1 l_1^2 \ddot{\theta}_1 + m_2 [l_1^2 \ddot{\theta}_1 + l_1 l_2 \{ \ddot{\theta}_2 \cos(\theta_1 - \theta_2) - \dot{\theta}_2 \sin(\theta_1 - \theta_2) (\dot{\theta}_1 - \dot{\theta}_2) \}] = -m_1 g l_1 \sin(\theta_1) - m_2 [l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2) + g l_1 \sin(\theta_1)]$$

$$\Rightarrow m_1 l_1^2 \ddot{\theta}_1 + m_2 l_1^2 \ddot{\theta}_1 + m_2 l_1 l_2 \ddot{\theta}_2 \cos(\theta_1 - \theta_2) - m_2 l_1 l_2 \dot{\theta}_2 \sin(\theta_1 - \theta_2) (\dot{\theta}_1 - \dot{\theta}_2) = -m_1 g l_1 \sin(\theta_1) - m_2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2) - m_2 g l_1 \sin(\theta_1)$$

$$\Rightarrow \ddot{\theta}_1 (m_1 l_1^2 + m_2 l_1^2) = -m_2 l_1 l_2 \ddot{\theta}_2 \cos(\theta_1 - \theta_2) + \underline{m_2 l_1 l_2 \dot{\theta}_2 \sin(\theta_1 - \theta_2) (\dot{\theta}_1 - \dot{\theta}_2)} - \underline{m_1 g l_1 \sin(\theta_1)} - \underline{m_2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2)} - \underline{m_2 g l_1 \sin(\theta_1)}$$

$$= -m_2 l_1 l_2 \ddot{\theta}_2 \cos(\theta_1 - \theta_2) + m_2 l_1 l_2 \dot{\theta}_2 \sin(\theta_1 - \theta_2) [\dot{\theta}_1 - \dot{\theta}_2 - \dot{\theta}_1] - g l_1 \sin(\theta_1) (m_1 + m_2)$$

$$= -\underline{m_2 l_1 l_2 \ddot{\theta}_2 \cos(\theta_1 - \theta_2)} - \underline{m_2 l_1 l_2 \dot{\theta}_2^2 \sin(\theta_1 - \theta_2)} - g l_1 \sin(\theta_1) (m_1 + m_2)$$

$$\Rightarrow \ddot{\theta}_1 (m_1 l_1^2 + m_2 l_1^2) = -m_2 l_1 l_2 [\ddot{\theta}_2 \cos(\theta_1 - \theta_2) + \dot{\theta}_2^2 \sin(\theta_1 - \theta_2)] - g l_1 \sin(\theta_1) (m_1 + m_2)$$

$$\Rightarrow \ddot{\theta}_1 = - \frac{m_2 l_2 [\ddot{\theta}_2 \cos(\theta_1 - \theta_2) + \dot{\theta}_2^2 \sin(\theta_1 - \theta_2)]}{l_1 (m_1 + m_2)} - \frac{g \sin(\theta_1)}{l_1}$$

